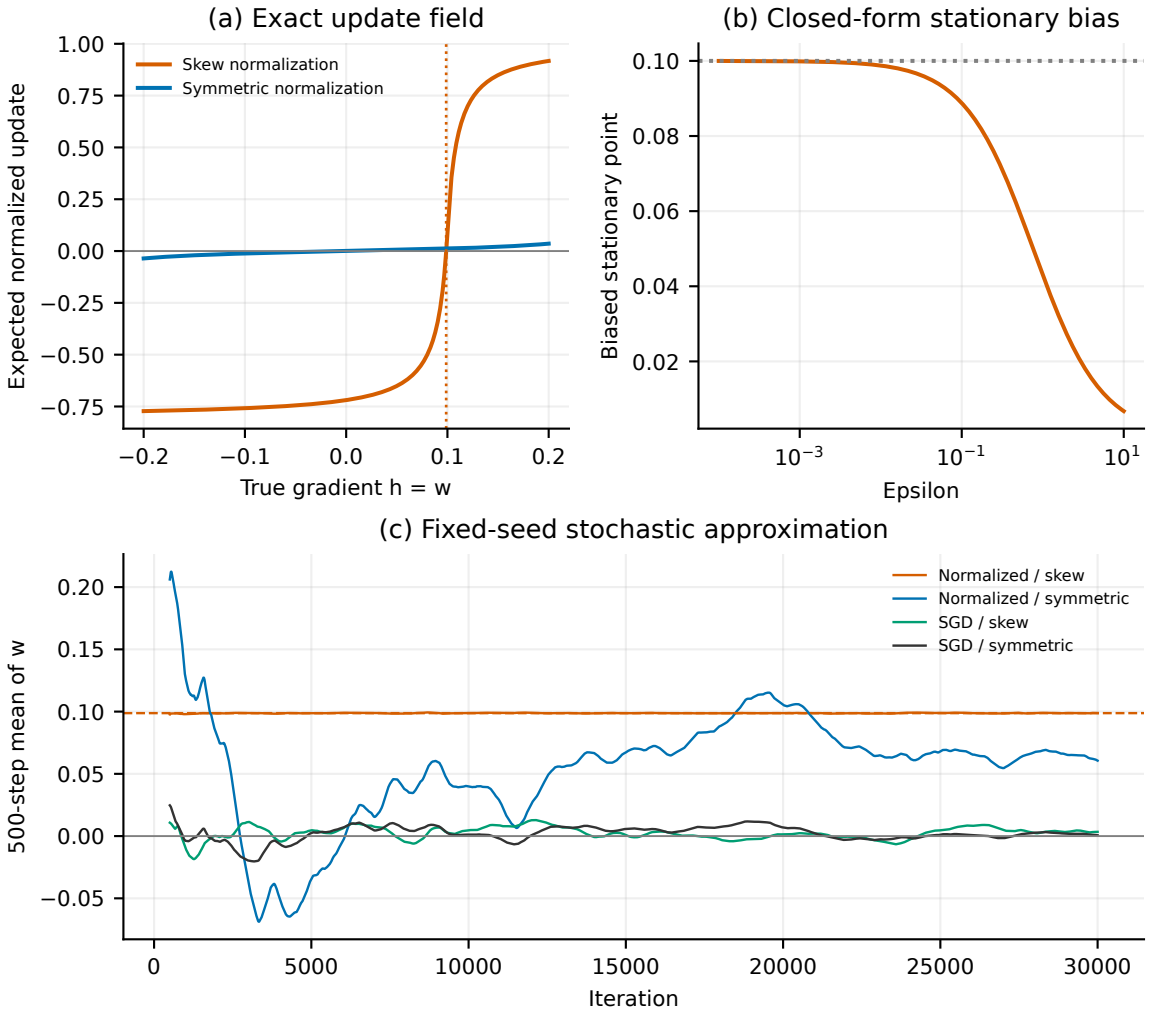
(a) Exact update field
Skew normalization
Symmetric normalization
Expected normalized update
True gradient h = w
(b) Closed-form stationary bias
Biased stationary point
Epsilon
(c) Fixed-seed stochastic approximation
Normalized / skew
Normalized / symmetric
SGD / skew
SGD / symmetric
500-step mean of w
Iteration